VICTORY 2027

The Circulatory System

Introduction to Circulatory System +

?Parts of Circulatory System (Blood, RBCs and WBCs)

LECTURE-2 BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. Open and closed circulatory system
2. Fluids inOur Body
3. Properties of Blood : The Blood
4. White blood cells (WBCs)
5. Origin and life of WBCs

Topics To Be Covered

6. Functions of Leukocytes (WBCs) – Body defence
7. Inflammation
8. Formation of antibodies
9. Blood platelets (Thrombocytes)
10. Antibodies and Antitoxins


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上



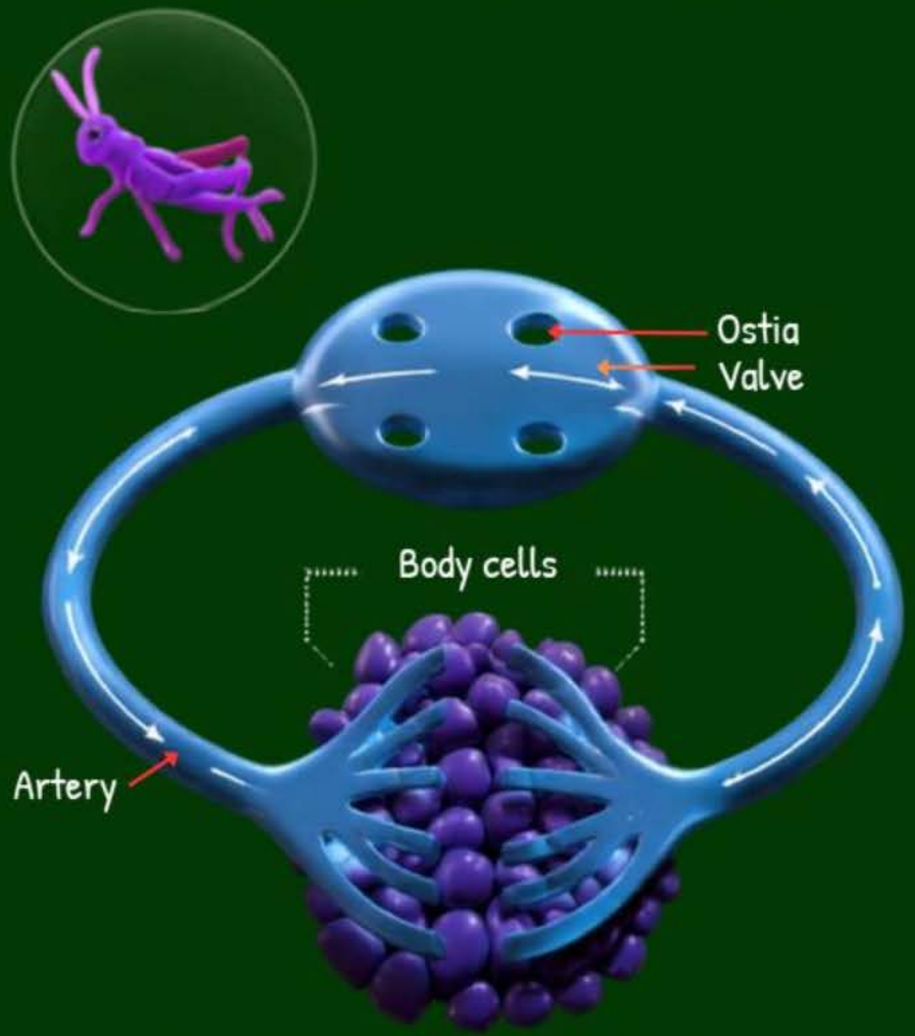
OPEN AND CLOSED CIRCULATORY SYSTEM

Open Circulatory System

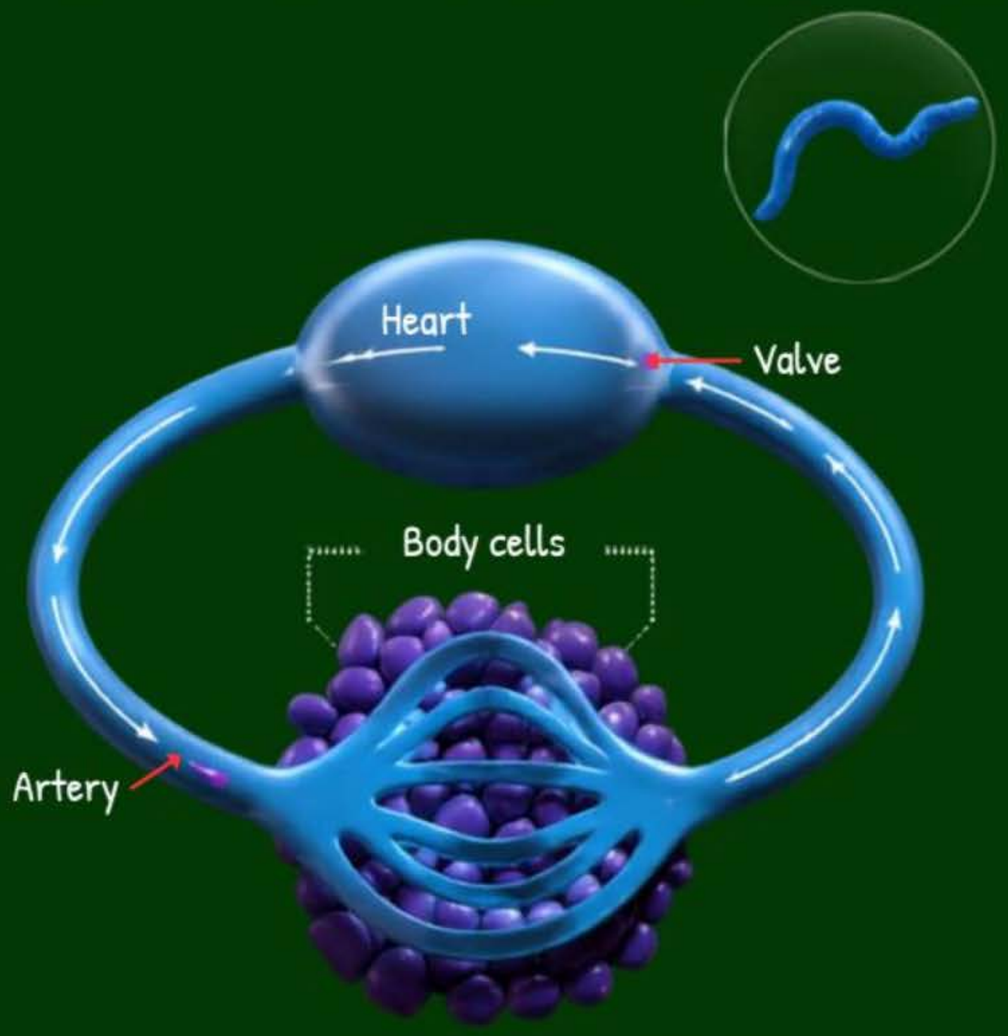
- Blood in Humans, squids, Cats, earthworms circulates in a closed manner i.e. all the time through blood vessels. Such a type of blood circulation is called a **closed blood circulatory system.**

Closed Circulatory System

- In animals such as Snails, clams, cockroaches and spiders the blood mostly flows through open spaces in the body, called an **open blood circulatory system.**
- Their blood flows from the heart to body spaces without vessels.

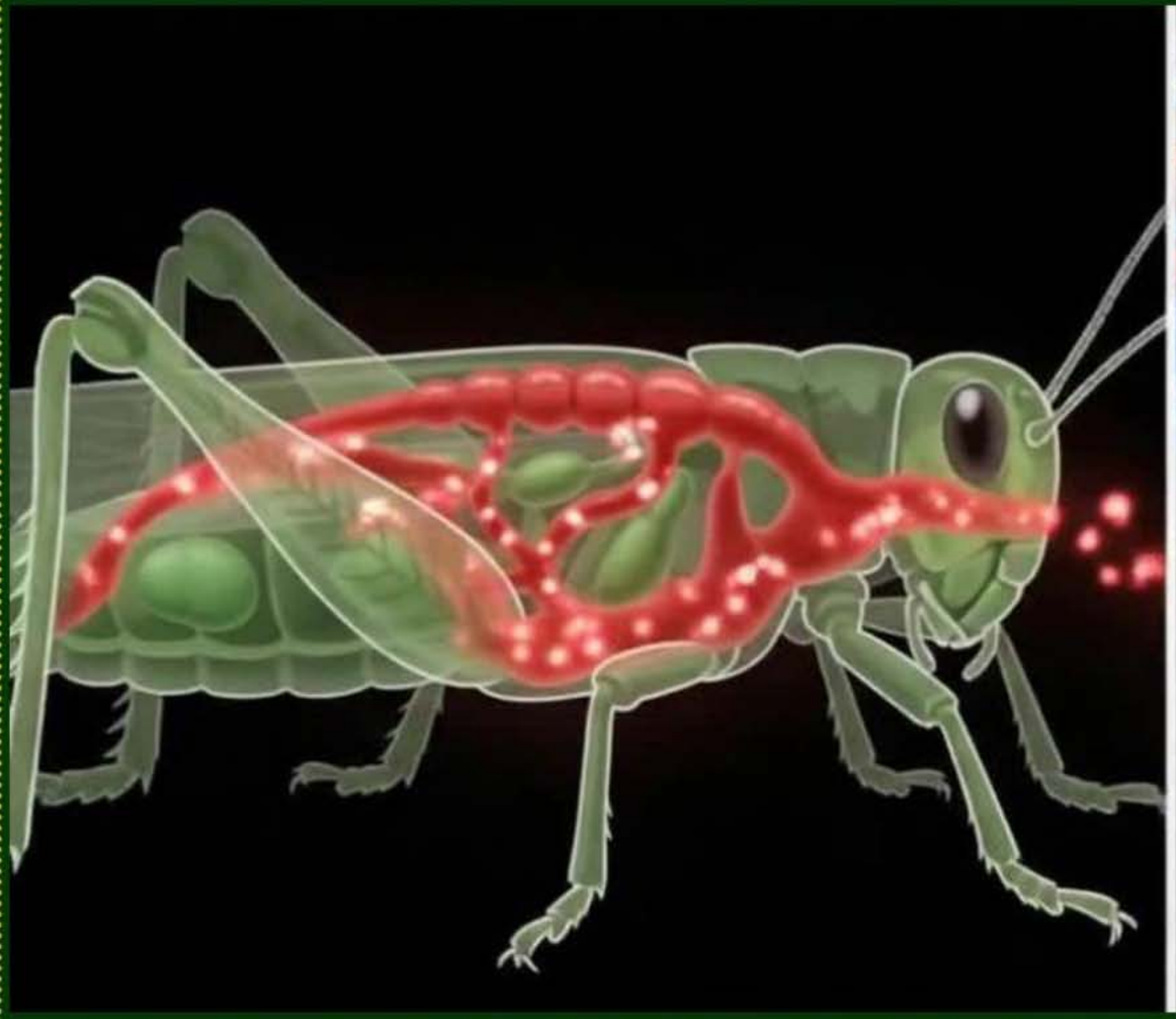


Open Circulatory System

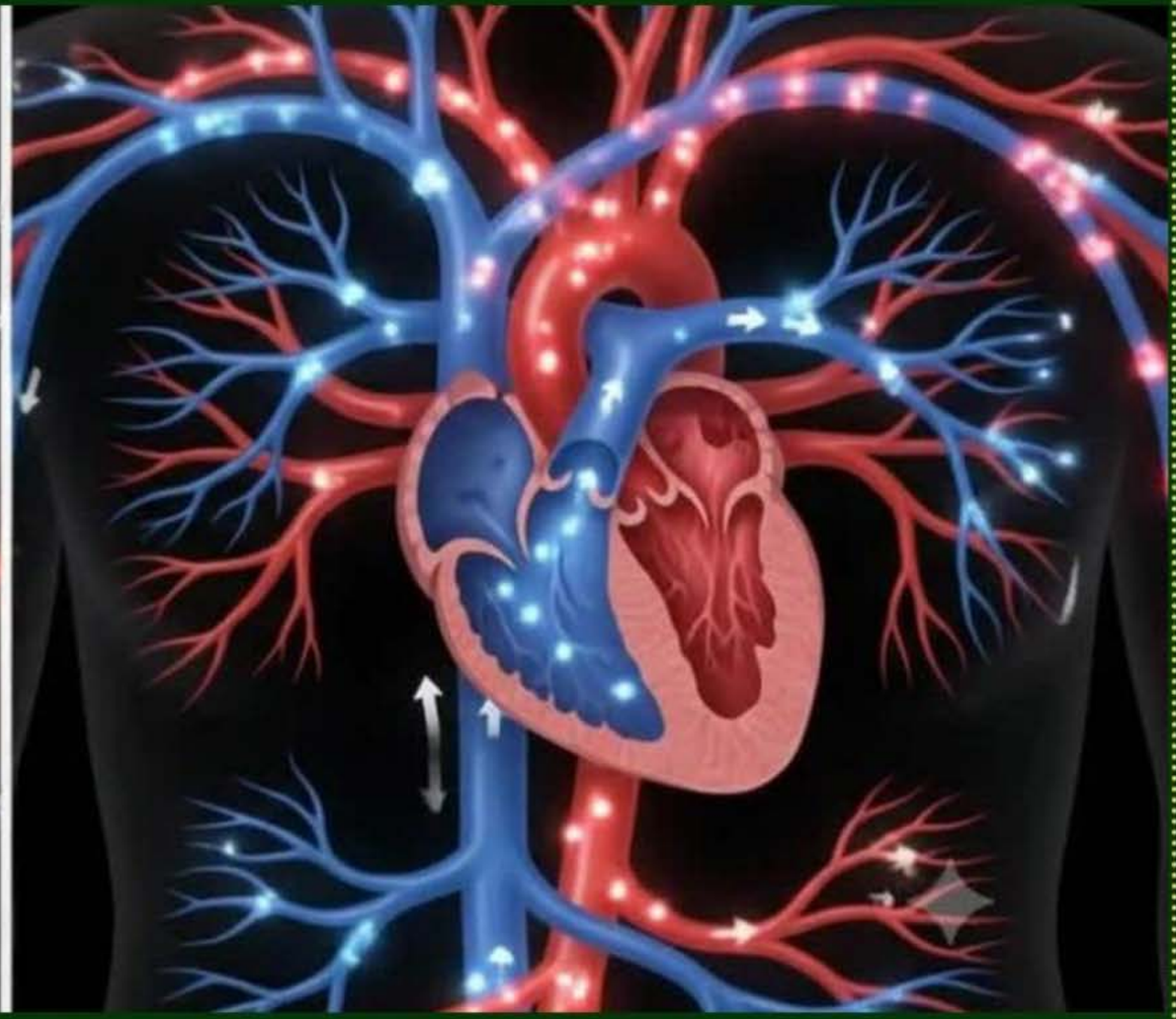


Closed Circulatory System

Open Circulatory System



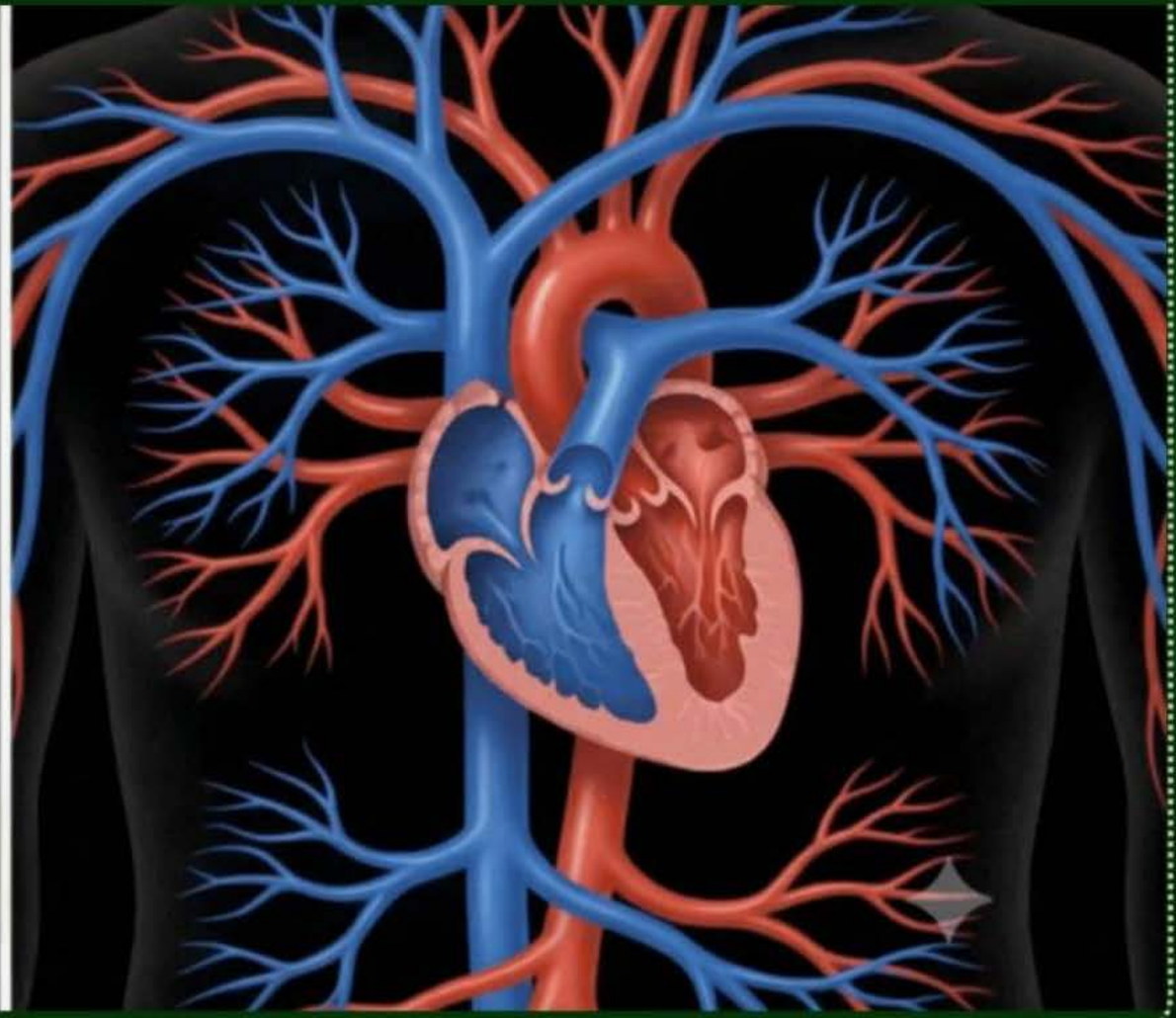
Closed Circulatory System



Open Circulatory System

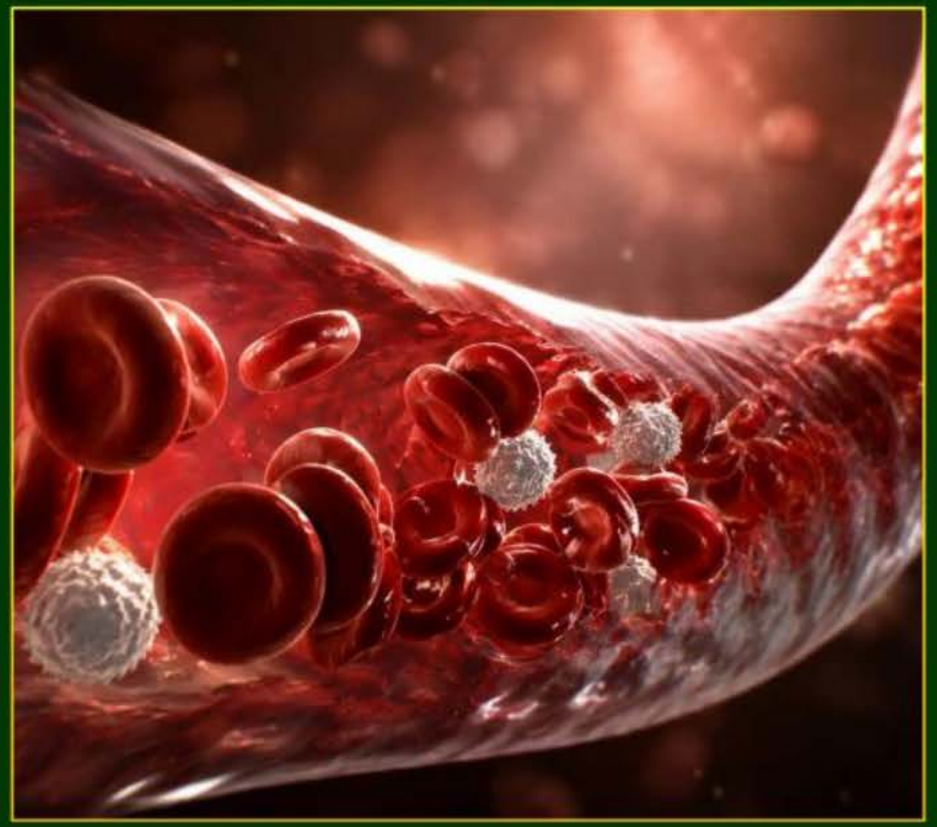


Closed Circulatory System



FLUIDS IN OUR BODY

- Blood, contained in the heart and in the blood vessels (arteries, veins and capillaries) of the circulatory system.
- Tissue fluid, occupying spaces between cells in the organs.
- Lymph, which is contained within lymph vessels and lymphatic organs such as the spleen and the tonsils.



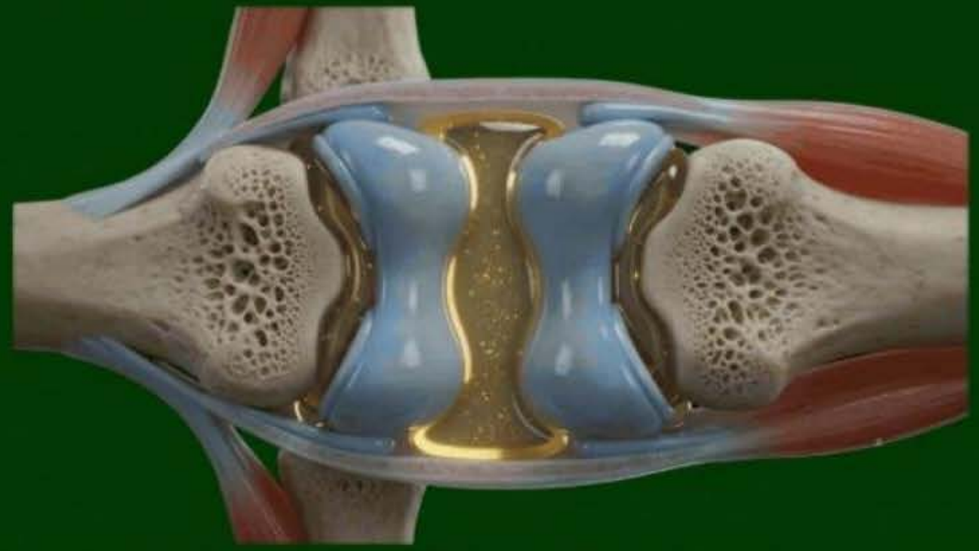
NON CIRCULATING FLUIDS IN OUR BODY

- Some fluids located in particular organs such as synovial fluid filled in the cavities of skeletal joints, vitreous humour in the eye, etc. They do not circulate.



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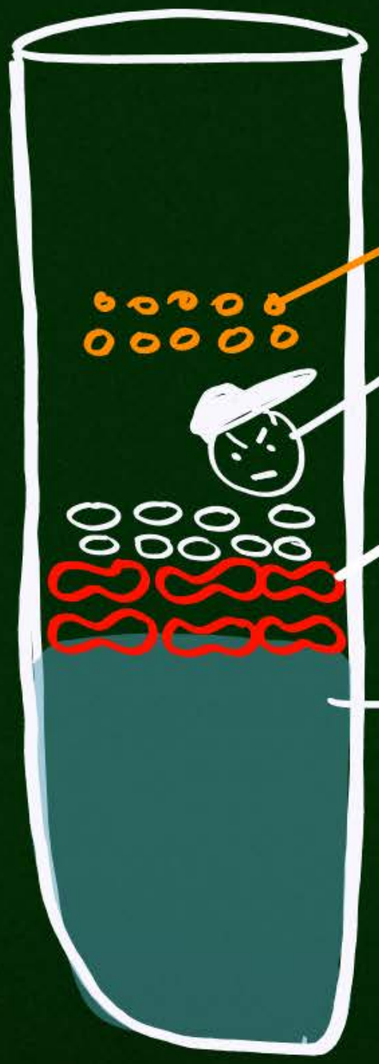




PROPERTIES OF BLOOD : THE BLOOD

- **Never Stationary –** Blood is always in motion from the heart to the arteries and back through the veins.
- **Colour -** The blood is a somewhat thick fluid, bright red when taken from an artery or dark red when taken from a vein.
- **Volume –** An average adult human has 5 to 6 litres of blood by volume in his body.
- **Taste –** saltish – The blood is slightly alkaline with a pH of 7.3 to 7.45 (7 is neutral, neither acidic nor alkaline).

Composition of Blood



Anti coagulant

HEPARIN

↓

Does not allow blood to clot.

CORPUSCLES / ELEMENTS / CELLS] 45%

[2-4 lakh] / mm^3

- **Platelets** [Thrombocytes]
- **WBCs** [Leucocytes] [4-5000] / mm^3 (White)
- **RBCs** [Erythrocytes] per mm^3 (Red cell)

[4-5 million] / μl

PLASMA 55%

Matrix of Blood

- Water [90-92%]
- Proteins [8%]
- Hormones, Ions, etc. [1-2%]

PYQ.

Composition of Plasma is _

A) 40-45%.

B) 50-55%.

C) 60-70%.

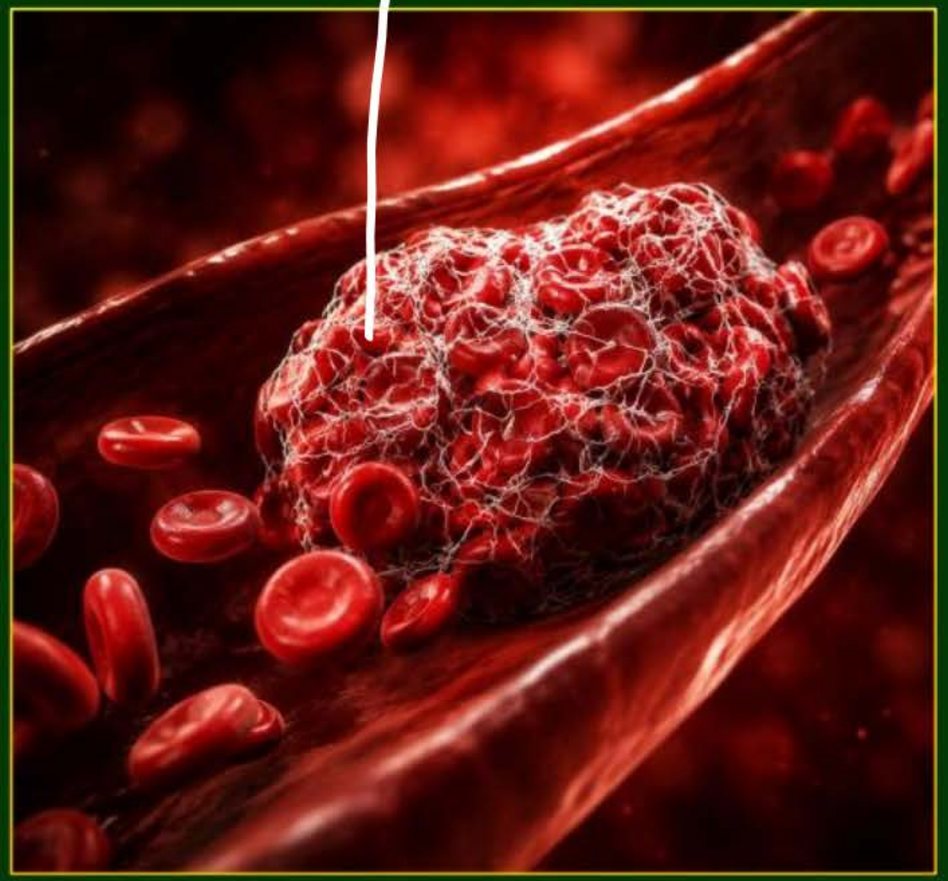
D) 40-50%.

FUNCTIONS OF BLOOD

1. Blood forms a clot wherever there is a cut in a blood vessel. The clot serves to prevent

(i) Further loss of blood and

(ii) The entry of disease-causing germs.



PROTECTION BY BLOOD

1. Its white blood corpuscles protect the body from diseases by engulfing bacteria which may have entered the body.

1. It produces antitoxins and antibodies which neutralise the poisonous substances or kill the germs which enter the body.

COMPOSITION OF BLOOD

The blood consists of :

(i) Plasma - fluid part, constitutes 55-60 per cent of blood.

(ii) Cellular elements - red and white cells, and platelets, 40-45 per cent of blood.

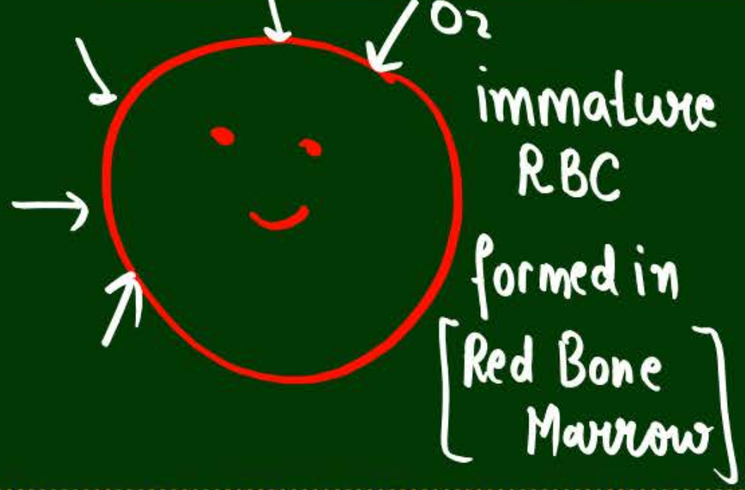
RED BLOOD CELLS (RBCS)

- RED BLOOD CELLS (RBCs) - The oxygen carriers



- Red blood cells are also called erythrocytes (erythros: red)

- These are minute biconcave disc-like structures, flat in the centre and thick and rounded at the periphery.





Mature RBC

Enucleated [No Nucleus]

No Endoplasmic Reticulum (for flexibility)

No Mitochondria (to prevent Respiration in it)

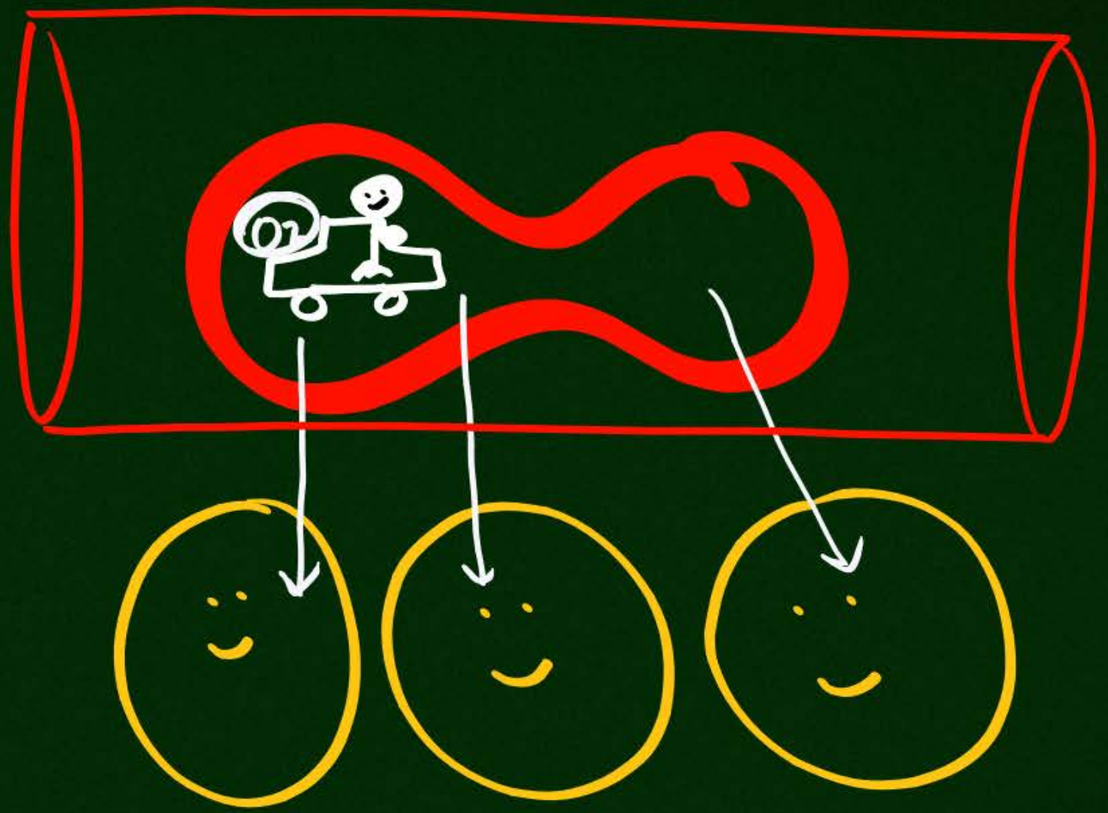
No Golgi Apparatus

Hb

$CO > O_2 > CO_2$



Biconcave / Discoid Shape





Haemoglobin (Hb)

Oxygen → O_2-Hb [Oxyhaemoglobin]

Oxygenated Blood.

Carbondioxide

Deoxygenated Blood

CO_2-Hb [Carbaminohaemoglobin]

[Carbon monoxide]
Poisonous Gas
[Hb.CO]
Carboxyhaemoglobin

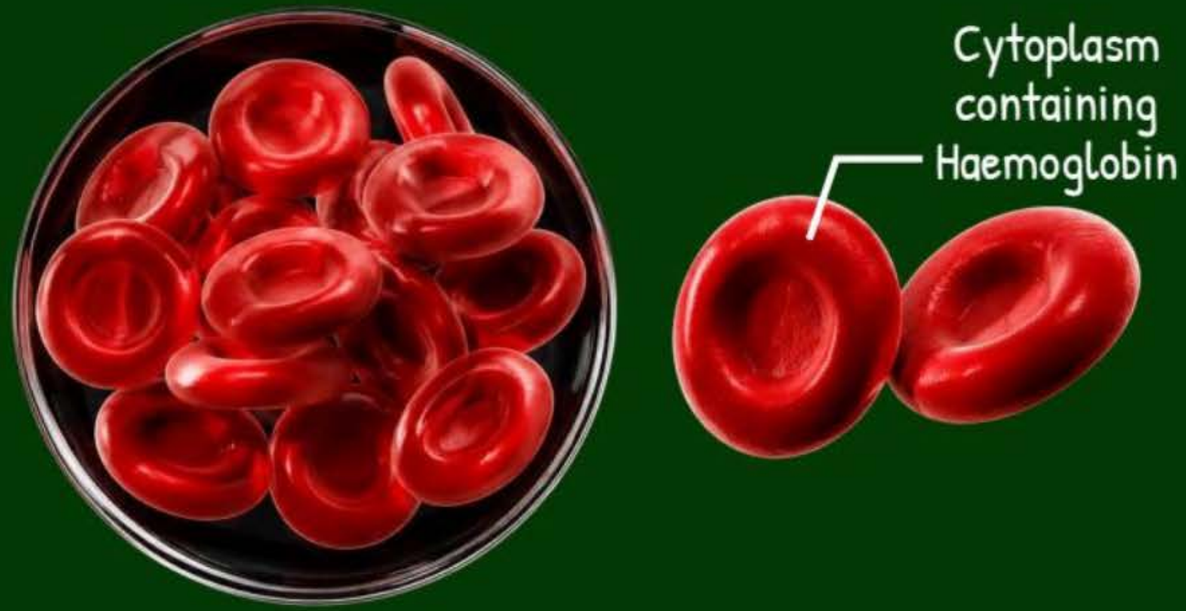
RBC

Smallest cell of body

- These are very small - about 7 micron in diameter (1 micron = μ one-thousandth of a millimetre and is represented by the symbol "μ".
- The small size provide a large surface area which makes them very efficient in absorbing oxygen.
- An adult human male has about 5 million RBC s per cubic mm of blood and an adult female has slightly less, about 4.5 million.

♂ Hb - 12-16 gm

♀ Hb - 10-12 gm.

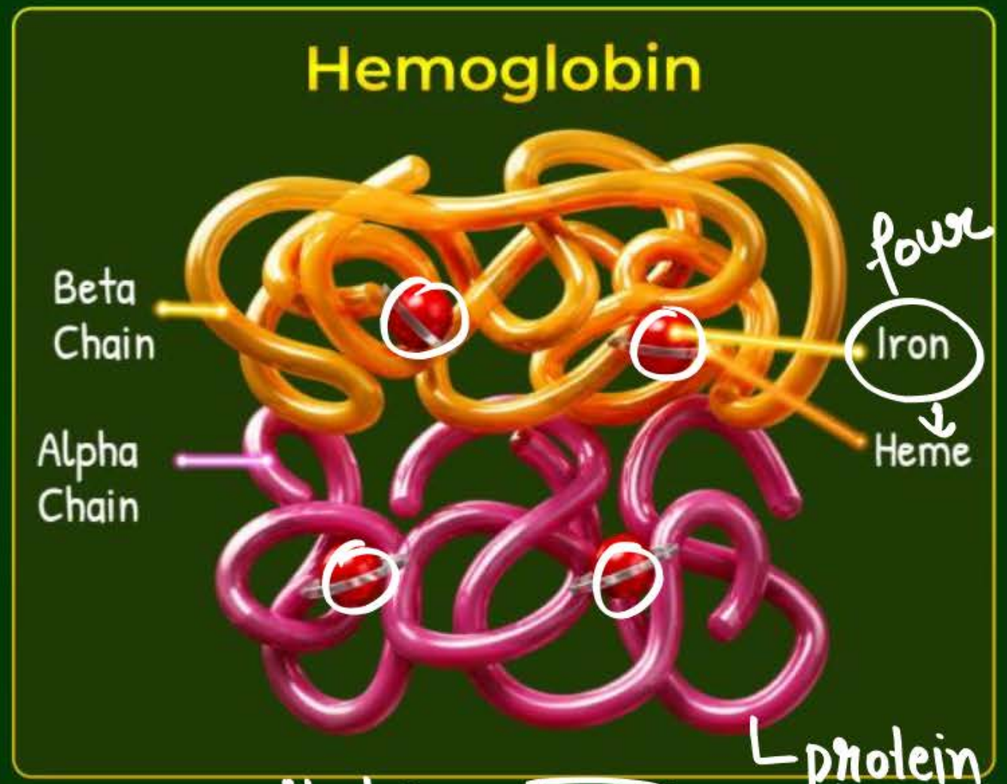


HAEMOGLOBIN



RBC
empty space STROMA + Hb
Hb or Hb of O2

- **HAEMOGLOBIN** is the effective chemical constituent of RBCs. The red blood cells have a colourless spongy body or stroma which contains a respiratory pigment haemoglobin (Hb).
- The haemoglobin is formed of an iron containing part (haemin) and a protein (globin).
- It has the ability to combine readily with oxygen to form oxyhaemoglobin, an unstable compound which readily gives up oxygen to the needy tissues.



Nahi बनाना

LIFE AND DEATH OF RBCS

- In adults, marrow of long bones, especially in the ribs, breast bone and ilium of hip girdle.



- In an embryo, in the liver and spleen.
- In children, in bone marrow of all bones until 5 years of age.

- The mature red blood cells are enucleated, but when they are being produced, they have one.

Q. Which of the following cells helps in defense?

A) Erythrocytes

B) Leucocytes

C) Platelets

D) None

Q. Which of the following cells are maximum in blood?

A) Erythrocytes
B) Leucocytes
C) Platelets
D) None

QUESTIONS

What would likely happen if blood did not clot after an injury?

A. Body temperature would drop

B. Pathogens would be transported faster

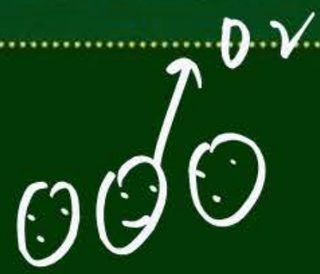
C. Uncontrolled blood loss would occur ✓

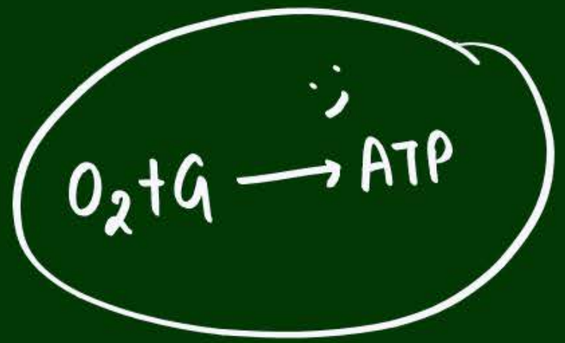
D. Oxygen transport would increase

QUESTIONS

How does oxyhaemoglobin assist in cellular respiration?

O_2 - Hb





A. Removes oxygen from tissues

B. Stores oxygen in lymph ✗

C. Prevents oxygen from dissolving in plasma

D. Releases oxygen at tissues for use in respiration

Thank You